Supplementary Material for Confidence-Shielded Evaluator-Conditioned Fusion for Risk-Controlled Infrared–Visible Object Detection

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1 Data provenance and split control

LLVIP uses a fixed 1,350/150/300 train/validation/test partition. The manifest contains the complete stem lists and SHA-256 digests. All three pairwise stem intersections are empty. The 300 evaluated images are the official test images; they are not used for checkpoint selection.

Table 1: Partition sizes and annotation counts.

Dataset	Training	Validation	Test	Test groups
LLVIP	1,350	150	300	7 scenes
M3FD	1,600	400	200	2 frame groups
DroneVehicle	public train	public validation	300	3 frame groups

Using the same training-run-cluster and group bootstrap for the primary CS-ECF-versus-visible contrast gives a dataset-balanced AP50 gain of 0.0133 and a 95% interval of $[0.0026, 0.0261]$. This clustered interval is the primary uncertainty statement; treating the two DroneVehicle checkpoints as independent would understate their shared training-run dependence.

Table 2: LLVIP annotation counts in the fixed split and pairwise overlap.

Split	Pedestrian boxes	Pairwise stem overlap
Training	4,479	train–validation: 0
Validation	419	validation–test: 0
Test	744	train–test: 0

M3FD frames are partitioned into consecutive 100-frame groups. The initial grouping seed is 2028 and the extension seed is 2029; the two held-out test groups do not enter training or validation. DroneVehicle uses three consecutive 100-frame groups selected from the official test archive with seed 2037. The source archive has 4,430,343,899 bytes and SHA-256 `94BF47E493E7A57FD5D900439F614F339E7B6C7181E4D781BCFEBAEB963FFD8E`.

The YOLO11x-OBB best and last checkpoints share one 150-epoch training run (batch size 32, 640-pixel input, seed 0). They are treated as one training-run cluster in every aggregate bootstrap.

2 Detection evaluation

All predictions are generated at a confidence floor of 0.001 and a 640-pixel inference size. YOLO predictions use NMS at IoU 0.7. For each image, class and IoU threshold from 0.50 to 0.95 in increments of 0.05, predictions are assigned to ground truth by one-to-one maximum-cardinality matching over all valid prediction–annotation pairs. The resulting true positives and confidence scores are pooled across the evaluated images, and AP is computed on the 101-point interpolated precision–recall curve with Ultralytics 8.4.189. AP50 averages class AP at IoU 0.50; AP50–95 additionally averages the ten IoU thresholds.

3 Complete CS-ECF and Always CDF results

The tables below reconstruct the frozen policy from released per-image detections. "Selected" gives the number of images assigned to visible, CDF, PIAFusion and SPGFusion. The group intervals are percentile bootstraps over scenes or consecutive frame groups.

Table 3: Per-checkpoint paired comparison on LLVIP. Delta is CS-ECF minus Always CDF; the last column is Visible/CDF/PIAFusion/SPGFusion selections.

Evaluator	CDF	CS-ECF	Delta	95% group CI	V/C/P/S
YOLO11n	.8920	.8845	−.0076	[−.024, +.012]	276/1/9/14
Agnostic	.8629	.8611	−.0018	[−.022, +.011]	286/2/6/6
YOLO26n s0	.8200	.8088	−.0112	[−.034, +.009]	288/1/3/8
RT-DETR-l	.8100	.8136	+.0036	[−.015, +.018]	293/0/0/7
YOLO26n s1	.8048	.7911	−.0138	[−.026, +.002]	265/1/4/30
YOLO26n s2	.8183	.8087	−.0096	[−.023, +.012]	282/0/0/18
YOLOv8n s0	.7847	.7761	−.0086	[−.025, +.011]	282/1/0/17
YOLOv8n s1	.8087	.8119	+.0032	[−.015, +.024]	290/0/0/10

Table 4: Per-checkpoint paired comparison on M3FD and DroneVehicle.

Dataset	Evaluator	CDF	CS-ECF	Delta	95% group CI
M3FD	YOLO26n	.3705	.3757	+.0053	[−.0002, +.0114]
M3FD	YOLOv8n	.3731	.3972	+.0241	[+.0087, +.0321]
DroneVehicle	best	.6624	.6814	+.0190	[−.0013, +.0301]
DroneVehicle	last	.6524	.6767	+.0243	[−.0067, +.0464]

Table 5: Paired CS-ECF minus Always CDF summary.

Scope	Difference	95% interval
LLVIP	−0.0057	[−0.0189, +0.0156]
M3FD	+0.0147	[+0.0100, +0.0159]
DroneVehicle	+0.0217	[−0.0007, +0.0738]
Dataset-balanced	+0.0102	[−0.00057, +0.02980]

The combined interval resamples scene or frame groups and training-run clusters within each dataset, then assigns equal weight to the three datasets. There are eight run clusters on LLVIP, two on M3FD and one on DroneVehicle. The DroneVehicle component therefore quantifies test-group uncertainty but cannot estimate variability between independent DroneVehicle training runs. The lower endpoint of the combined interval is below zero, so the direct CDF comparison does not establish a separated average improvement.

4 External evaluator stress tests

Two COCO-pretrained detectors, YOLOv9t and YOLOv10s, were used without LLVIP fine-tuning. They were not used in policy construction, threshold selection, or the primary evaluation. Person-class detections were retained before AP and selection scoring. The 1.5 margin and top-five score were unchanged.

A third stress test used an LLVIP-fine-tuned YOLOv9t. The checkpoint was initialized from COCO YOLOv9t and trained on the 1,350 visible training images for a pre-specified 40 epochs. Training used AdamW, initial learning rate 0.002, three-epoch warm-up, cosine decay, weight decay 5×10^{-4}, batch size 16, 640-pixel inputs, random horizontal flipping, and brightness/contrast jitter. A reproducible PIL-based loader calling the Ultralytics detection loss was used. The final checkpoint

after epoch 40 was frozen before the LLVIP test images were scored; no epoch was selected using test results. Its SHA-256 is `AE0A44359C489DEEAE3F7443173588B9F68E264B6BE1EA455E961EB2DED0797D`. The checkpoint reached mAP50 0.8456 on the 150-image validation set and 0.6666 on the 300-image test visible input.

Table 6: External evaluator stress tests on the same 300 LLVIP test pairs.

Model	Visible	Always CDF	No shield	CS-ECF
YOLOv9t, zero-shot	.7698	.7888	.8216	.7948
YOLOv10s, zero-shot	.8344	.8617	.8699	.8557
YOLOv9t, LLVIP-trained	.6666	.6598	.6405	.6558

Table 7: Paired confidence intervals and scene-tail behavior for the external evaluators. The final column reports the worst scene change for CS-ECF / no-shield.

Model	CS-ECF − visible	CS-ECF − CDF	Worst (CS/NS)
YOLOv9t, zero-shot	$+.0250$ $[+.0053, +.0523]$	$+.0060$ $[-.0277, +.0374]$	$-.0167/-.0011$
YOLOv10s, zero-shot	$+.0213$ $[+.0041, +.0439]$	$-.0059$ $[-.0266, +.0180]$	$-.0038/-.0337$
YOLOv9t, LLVIP-trained	$-.0108$ $[-.0342, +.0050]$	$-.0040$ $[-.0206, +.0108]$	$-.0724/-.1208$

The shield selected visible on 233/300 zero-shot YOLOv9t, 245/300 YOLOv10s, and 230/300 LLVIP-trained YOLOv9t images. The two zero-shot evaluators support a positive difference from visible but not superiority over Always CDF. The fine-tuned YOLOv9t is a negative portability case: CS-ECF was below both visible and Always CDF, with scene-group intervals crossing zero. The shield improved the unshielded selector by 0.0153 AP50 (group interval $[+0.0072, +0.0267]$) and reduced the worst scene loss from 0.1208 to 0.0724, but did not remove the loss. Tail behavior therefore remains evaluator dependent rather than universally controlled.

5 Validation choice and exploratory alternatives

The policy was fixed before the confirmatory comparisons. On validation, the four-candidate top-five selector with a 1.5 visible margin produced gains of 0.001850 and -0.000070 for the two LLVIP evaluators and 0.001518 and 0.000083 for the two M3FD evaluators. Its two-dataset balanced validation average was 0.000845. The candidate set was retained to preserve the visible, region-selected, learned, and published fusion families; the margin was fixed at 1.5.

Two alternative policies were evaluated after the frozen comparison. They are exploratory and are not used to support the main claim. A top-10 selector over visible/CDF/PIAFusion with margin 1.3 gave a 0.0039 balanced gain but a hierarchical interval of $[-0.0029, +0.0127]$. A mean-confidence selector over visible/SPGFusion with margin 1.5 gave a -0.0034 balanced gain and $[-0.0085, +0.0013]$. Both alternatives fail to establish a stable advantage across the 10 LLVIP/M3FD checkpoints.

6 Thermal controls

Table 8: Spearman association between switching and thermal or visible features. These secondary association tests are descriptive and are not multiplicity-adjusted.

Dataset	Thermal top-hat ρ (p)	Visible luma ρ (p)	CDF-mask ρ (p)
LLVIP	$+0.151$ (0.009)	-0.104 (0.071)	-0.029 (0.614)
M3FD	$+0.320$ ($< 10^{-5}$)	-0.066 (0.356)	-0.332 ($< 10^{-5}$)
DroneVehicle	-0.403 ($< 10^{-5}$)	-0.441 ($< 10^{-5}$)	$+0.064$ (0.269)

In the fixed-input CDF ablation on DroneVehicle, the thermal mask changed AP50 from 0.6620 to 0.6624 for the best checkpoint and from 0.6512 to 0.6524 for the last checkpoint, relative to global detail injection.

The association reverses between datasets. Thermal contrast is therefore not treated as a sufficient proxy for fusion utility, and no universal thermal-switching rule is claimed.

7 Reproduction

The public repository contains the selector, candidate-generation and detector scoring code, per-image score arrays, split manifests, source data for every figure and the scripts that reproduce the intervals. The direct baseline comparison is generated by:

```
python scripts/analyze_csecf_vs_cdf.py
```

This command validates each checkpoint AP before writing `results/selective_fusion/csecf_vs_cdf_paired.json`. Split and candidate manifests retain their seeds and selected groups. The independent evaluator checkpoints were trained or loaded by `scripts/train_independent_manual.py`, scored with `scripts/score_clean_evaluators.py`, and summarized with `scripts/analyze_independent_evaluators.py`. The LLVIP-trained YOLOv9t result is stored in `results/independent_evaluators/domain_yolov9t_v3/`.